

# Coursework Assignment Brief

## Assessment 2 - Undergraduate

### Academic Year 2025-26 S1

|                                                                      |                                    |                                        |                                                                                  |
|----------------------------------------------------------------------|------------------------------------|----------------------------------------|----------------------------------------------------------------------------------|
| Module Title: Artificial Intelligence and Machine Learning           |                                    |                                        |                                                                                  |
| Module Code: CMP5367                                                 |                                    |                                        |                                                                                  |
| <b>Assessment Type</b><br>D2: Presentation (In Person)               | <b>Level</b><br>5                  | <b>Weighting</b><br>20%                | <b>Word Count</b><br>10 Slides                                                   |
| <b>Submission Date</b><br>Presentation:<br>December 15 <sup>th</sup> | <b>Submission Time</b><br>03:00 PM | <b>Module Leader</b><br>Hadeel Saadany | <b>Time Limit (for in person presenting and oral assessments):</b><br>20 minutes |

#### Assessment Brief:

| Assessment Information |                                                                                               |
|------------------------|-----------------------------------------------------------------------------------------------|
| Assessment Summary     | D2 Part 1 [Group]: PowerPoint Slides (10%)<br>D2 Part 2 [Group]: In-person Presentation (10%) |
| Assessment Title       | Artificial Intelligence using Machine Learning Models                                         |
| Things to include:     | See description of the task further below in this document.                                   |

|                                                                             |                                                                                                                                          |
|-----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Completion of this assessment will address the following learning outcomes: |                                                                                                                                          |
| 1                                                                           | Compare the different aspects involved in the modern Artificial Intelligence, Machine Learning and Data Science.                         |
| 2                                                                           | Critically evaluate and practice a range of Machine Learning algorithms, Data Science tools, and frameworks for developing AI solutions. |
| 3                                                                           | Apply the learned algorithms, tools and frameworks to solve real life problems.                                                          |
| 4                                                                           | Demonstrate skills in formulating research problems and presenting them.                                                                 |

#### Submission Information

Complete the appropriate cover sheet for this assessment and append your work.

Submit this coursework assessment task via Moodle.

### **Late Submission**

Assessments must be submitted in the format specified in the assessment task, by the deadline and to the submission point published on Moodle. Failure to submit by the published deadline will result in penalties which are set out in Section 6 of the Academic Regulations, available at:

<https://icity.bcu.ac.uk/Quality-Enhancement-and-Inclusion/Quality-Assurance-and-Enhancement/Academic-Regulations>

### **Word Count**

The maximum word count for this module assessment is shown on Page 1. A +10% margin of tolerance is applied, beyond which nothing further will be marked. Marks cannot be awarded for any learning outcomes addressed outside the word count.

The word count refers to everything in the main body of the text (including headings, tables, citations, quotes, lists etc.). Everything before (i.e. abstract, acknowledgements, contents, executive summaries etc.) and after (i.e. references, bibliographies, appendices etc) is **not** included in the word count limit.

### **Referencing Style**

- BCU Harvard

More information on referencing is available here: <https://www.bcu.ac.uk/library/services-and-support/referencing>

### **Use of Artificial Intelligence**

Whilst AI tools can be helpful in assisting learning, when it comes to assessment, the Academic Misconduct Procedure is clear that this should be a student's own original work and not the work of other people or AI tools.

The [Use of AI Tools – Student Guidelines](#) document follows the same guidelines your lecturers use. If you are unsure of whether AI is appropriate within your work, please read the guidelines or ask your lecturer. For advice and guidance around academic writing, please visit the [Centre for Academic Success](#).

### **Academic Integrity Guidance**

Academic integrity is the attitude of approaching your academic work honestly, by completing and submitting your own original work, attributing and acknowledging your sources when necessary. Understanding good academic practice in written and oral work is a key element of academic integrity. It is a positive aspect of joining an academic community, showing familiarity with and acknowledging sources of evidence. The skills you require at higher education may differ from those learned elsewhere such as school or college.

You will be required to follow specific academic conventions which include acknowledging the work of others through appropriate referencing and citation as explicitly as possible. If you include ideas or quotations that have not been appropriately acknowledged, this may be seen as plagiarism

which is a form of academic misconduct. If you require support around referencing, please contact the [Centre for Academic Success](#)

It is important to recognise that seeking out learning around academic integrity will help reduce the risk of misconduct in your work. Skills such as paraphrasing, referencing and citation are integral to acting with integrity and you can develop and advance these key academic skills through the [Centre for Academic Success \(CAS\)](#).

To learn more about academic integrity and its importance at university, you can access CAS resources on Moodle. Furthermore, you can book on to workshops and request 1-2-1 support around key academic skills.

### **Academic Misconduct**

Academic misconduct is conduct that has or may have the effect of providing you with an unfair advantage by relying on dishonest means to gain advantage and which therefore compromises your academic integrity.

The Academic Misconduct procedure sets out the process we will follow, and the penalties we may apply, in cases where we believe you may have compromised your academic integrity by committing academic misconduct. The Academic Misconduct Procedure and information about academic support is available at: <https://icity.bcu.ac.uk/Student-Affairs/Appeals-and-Resolutions/Academic-Misconduct-Procedure>

**Title:** Build and evaluate a machine learning model to solve a real-life problem (e.g., an industrial challenge) using an appropriate dataset.

**Style:** In Person Presentation.

**Rationale:** This assessment is most suited for assessing the learning outcomes of the module providing the practical nature of the AI field. The presentation ensures the student's ability to demonstrate an understanding of the dataset analysis and machine learning algorithm. It is intended to show how the student can answer queries on the AI problem and experiment tackled in assessment 1 of this course.

**Description:**

You are required to present your findings as well as the experiment that were submitted as part D1.a, the Machine Learning Portfolio of assessment 1 for this module. You will present your chosen dataset attributes, preprocessing, analysis and model training. You will also provide a demonstration of the results and your conclusion on the whole experiment.

**Additional information:**

**D2 - Presentation (20%) [PowerPoint Slides (10%), In-person Presentation (10%)]**

1. You are required to create a PowerPoint presentation that describes your work in the machine learning portfolio you submitted as part of the deliverable 1 (D1.a Machine Learning Portfolio). Ensure your slides are clear, concise, and well-organized.
2. Ensure your slides are visually appealing and include appropriate graphics, charts, or tables to support your explanations.
3. Be prepared to present your slides and answer questions on the content. Your answers should show your understanding of the dataset and the machine learning experiment you have conducted in your individual report.

**NB:**

Use these guidelines and instructions as a template to create your presentation, focusing on understanding the dataset in your machine learning projects. This structured approach will help ensure that you comprehensively address each step and provide a clear and organized explanation of the dataset.

***Introduction***

- **Purpose:** State the objective of your presentation. This should be brief and provide a roadmap of what will be covered.

***Dataset Overview:***

- **Title:** Introduce the dataset with a concise title that describes its content and relevance to your project.

- **Relevance:** Explain why this dataset is important for your machine learning project.

### **Data Source**

- **Origin:** Describe where the dataset was obtained from and discuss its reliability and relevance.

### **List Attributes**

- **Attributes:** List and describe each attribute (column) in your dataset.
  - **Details:** Include the attribute name, data type (categorical, numerical), and a brief explanation of what it represents.
  - **Structure:** Provide an understanding of the dataset's structure.

### **Data Preprocessing**

Address the following preprocessing tasks (if any is applicable):

- **Handling Missing Values:** Explain your approach to dealing with missing values, such as imputing or removing them.
- **Encoding Categorical Variables:** Describe how you will convert categorical variables into numerical format, mentioning methods like one-hot encoding or label encoding.
- **Scaling Numerical Attributes:** Detail why scaling numerical attributes is necessary and explain the method you will use (e.g., standardization or normalization).
- **Balancing the Target Variable:** Discuss techniques like oversampling or under-sampling to handle class imbalance in the dataset.

### **Machine Learning Algorithms and Hyperparameter Tuning**

- **Algorithms:** Mention the machine learning algorithms you used for your project.
- **Hyperparameter Tuning:** Discuss any strategies you employed for hyperparameter tuning to optimize model performance.

### **Potential Challenges**

Highlight challenges or potential issues that affected your machine learning project. Demonstrate your awareness of these obstacles:

- **Imbalanced Dataset:** Discuss how you addressed class imbalance.
- **Handling Categorical Variables:** Explain the challenges and your strategy for encoding categorical variables.
- **Feature Selection:** Mention how you selected the most relevant features, using techniques such as feature importance or selection algorithms.

### **Conclusion**

- **Summary:** Recap the key points from your presentation and reiterate the importance of understanding the dataset.

- **Future Steps:** Emphasize that this report serves as a roadmap for subsequent stages of your project, including data exploration, modelling, and evaluation.

For advice on writing style, referencing and academic skills, please make use of the Centre for Academic Success: [Centre for Academic Success - student support | Birmingham City University \(bcu.ac.uk\)](https://www.bcu.ac.uk/academic-success)

**Transferable skills:**

- Problem solving
- Time keeping
- Presentation skills.
- Project management
- Written communication skills



## D2 [Individual]: PowerPoint Slides - In-person Presentation – Table of Marking Criteria (20%)

| Criterion                                                      | 0-20%<br>Fail              | 20-39%<br>Fail                                                          | 40-49%                                                                                                                                                            | 50-59%                                                                                                                                                      | 60-69%                                                                                                                                                                                                                                                 | 70-79%                                                                                                                                    | 80-100%                                                                                                                                                          |
|----------------------------------------------------------------|----------------------------|-------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Slides</b><br><br>(10% of overall module mark)              | No slides submitted.       |                                                                         | Slides covers only some of the required sections. The structure of the presentation is hard to follow (e.g., due to repetition of information or lack of clarity) | Slides cover all the required sections, but the structure of the presentation is hard to follow (e.g., due to repetition of information or lack of clarity) | Slides cover all the required sections. The structure of the presentation is logical in some parts only.<br>Slides cover all the required sections. The structure of the presentation is logical. Stronger arguments need to be made in some sections. | Slides cover all the required sections. The structure of the presentation is logical, and the arguments made for the solution are strong. | Slides cover all the required sections. The structure of the presentation is logical, and the arguments made for the solution are strong and backed by research. |
| <b>In-person presentation.</b><br>(10% of overall module mark) | Presentation not attended. | Presentation attended but did not present any part of the presentation. | The presenter exhibits low enthusiasm and confidence level. Questions are not well answered.                                                                      | Satisfactory dataset and justified questions, but some minor errors in initial analysis.                                                                    | Good choice of dataset and questions with fair impact and no errors in initial analysis.                                                                                                                                                               | Exhibits uneven enthusiasm and confidence level; Some question can be answered but not correct                                            | Excellent choice of dataset and questions are correctly answered, no errors in initial analysis.                                                                 |